Chapter 12 Dynamics of Relativistic Particle and Electromagnetic Fields

1. Motion of a Charged Particle in Electromagnetic Field

The covariant equation of motion for a charged particle in electromagnetic fields is

$$\frac{dp^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} U_\beta \qquad \Longrightarrow \qquad \begin{cases} \dfrac{d\vec{p}}{dt} &= q\left(\vec{E} + \dfrac{\vec{u}}{c} \times \vec{B}\right) \\ \dfrac{dp_0}{dt} &= \dfrac{q}{c}\,\vec{u}\cdot\vec{E} \end{cases} \tag{1}$$

where $p^\alpha = (mU_0,\, m\vec{U}\,)$, $U^\alpha = (\gamma c\,,\, \gamma\vec{u}\,)$, $\gamma = 1/\sqrt{1-(\vec{u}\cdot\vec{u})/c^2}$, and $\vec{u}$ is the velocity of the particle. To solve the motion of a charged particle, in many cases, it is convenient to transfer to a reference frame K' where the electromagnetic field is in one of three simplest configurations (a) $\vec{E} = 0$, (b) $\vec{B} = 0$, or (c) $\vec{E} \parallel \vec{B}$. Because of the invariances of the electromagnetic field with the Lorentz transformations,

$$\vec{E}\cdot\vec{B} = \text{invariant} \qquad \text{and} \qquad E^2 - B^2 = \text{invariant}\,,$$

those three simplest configurations of the electromagnetic field correspond to the following three situations,

In K Frame	Lorentz transformation	In K' Frame	
$\vec{E} \perp \vec{B}$ and $E < B$	$\Longrightarrow$	$E' = 0$	HW 12.5a
$\vec{E} \perp \vec{B}$ and $E > B$	$\Longrightarrow$	$B' = 0$	HW 12.5b
$\vec{E}\cdot\vec{B} = EB\cos\theta$ with $0 < \theta < 90°$	$\Longrightarrow$	$\vec{E}' \parallel \vec{B}'$	HW 12.6

2. Relativistic Lagrangian and Hamiltonian

With the Lagrangian $\mathcal{L}$ of a particle, the canonical momentum of the particle is defined as

$$\vec{p} = \frac{\partial\mathcal{L}}{\partial\vec{u}} \tag{2}$$

where $\vec{u} = \dot{\vec{x}}$ is the velocity of the particle. For a free particle, the relationship between the momentum and velocity should be consistent with the definition of the 4-momentum,

$$p^\alpha = (p_0\,,\, \vec{p}\,) = (\gamma mc\,,\, \gamma m\vec{u}\,)$$

The relativistic Lagrangian $\mathcal{L}_0$ of a free particle can thus be obtained by solving the differential equation,

$$\frac{\partial\mathcal{L}_0}{\partial u_i} = \frac{mu_i}{\sqrt{1-(u_1^2+u_2^2+u_3^2)/c^2}} \qquad \text{where } \ i = 1, 2, 3$$

which yields

$$\mathcal{L}_0 = \int du_i \frac{mu_i}{\sqrt{1-(u_1^2+u_2^2+u_3^2)/c^2}} = -mc\sqrt{c^2-(u_1^2+u_2^2+u_3^2)}$$
$$= -mc\sqrt{u_0^2-\vec{u}\cdot\vec{u}} = -mc\sqrt{u_\alpha u^\alpha} \qquad (3)$$

where $u^\alpha = (u_0\,,\,\vec{u}\,) = (c\,,\,\vec{u}\,)$. With the Lagrangian in Eq. (3), the relativistic 4-momentum can be obtained as

$$\begin{cases} p_0 = -\dfrac{\partial \mathcal{L}_0}{\partial u_0} = \gamma mc \\ \vec{p} = \dfrac{\partial \mathcal{L}_0}{\partial \vec{u}} = \gamma m\vec{u} \end{cases} \implies p^\alpha = -\frac{\partial \mathcal{L}_0}{\partial u_\alpha} \qquad (4)$$

where $u_\alpha = g_{\alpha\beta}u^\beta = (u_0\,,\,-\vec{u}\,)$.

For a charged particle in electromagnetic field, the Lagrangian in the Newtonian mechanics is

$$\mathcal{L} = \frac{1}{2}m(\vec{u}\cdot\vec{u}\,) + \frac{q}{c}\left(\vec{u}\cdot\vec{A} - u_0\Phi\right)$$

where $\mathcal{L}_0 = m(\vec{u}\cdot\vec{u}\,)/2$ is the Lagrangian of a free particle. The relativistic Lagrangian of a charged particle in electromagnetic field can be constructed similarly as

$$\mathcal{L} = -mc\sqrt{u_0^2-\vec{u}\cdot\vec{u}} + \frac{q}{c}\left(\vec{u}\cdot\vec{A} - u_0\Phi\right) = -mc\sqrt{u_\alpha u^\alpha} - \frac{q}{c}\,u_\alpha A^\alpha \qquad (5)$$

where $A^\alpha = (\Phi, \vec{A}\,)$. The Lagrangian equation is

$$\frac{\partial \mathcal{L}}{\partial \vec{x}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \vec{u}}\right) = 0\,, \qquad \forall\;\; i = 1,2,3 \qquad (6)$$

where

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \vec{u}}\right) = \frac{d}{dt}\left(\gamma m\vec{u} + \frac{q}{c}\vec{A}\right) = \frac{d\vec{p}}{dt} + \frac{q}{c}\left[\frac{\partial \vec{A}}{\partial t} + \frac{d\vec{x}}{dt}\cdot\frac{\partial \vec{A}}{\partial \vec{x}}\right]$$
$$= \frac{d\vec{p}}{dt} + \frac{q}{c}\left[\frac{\partial \vec{A}}{\partial t} + (\vec{u}\cdot\nabla)\vec{A}\right]$$

$$\frac{\partial \mathcal{L}}{\partial \vec{x}} = \frac{q}{c}\left[\nabla(\vec{u}\cdot\vec{A}\,) - c\nabla\Phi\right] = \frac{q}{c}\left[\vec{u}\times(\nabla\times\vec{A}\,) + (\vec{u}\cdot\nabla)\vec{A}\right] - q\nabla\Phi$$

and

$$\frac{\partial \mathcal{L}}{\partial \vec{x}} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \vec{u}}\right) = -\frac{d\vec{p}}{dt} + q\left[-\left(\nabla\Phi + \frac{1}{c}\frac{\partial \vec{A}}{\partial t}\right) + \frac{\vec{u}}{c}\times(\nabla\times\vec{A}\,)\right]$$

Therefore, the Lagrange's equation with the relativistic Lagrangian in Eq. (5) yields the equation of motion in Eq. (1), where

$$\vec{E} = -\nabla\Phi - \frac{1}{c}\frac{\partial \vec{A}}{\partial t} \qquad \text{and} \qquad \vec{B} = \nabla\times\vec{A}$$

For a charged particle in electromagnetic field, the canonical momentum conjugate to the coordinate x^α is

$$P^\alpha = -\frac{\partial \mathcal{L}}{\partial u_\alpha} = mU^\alpha + \frac{q}{c}A^\alpha \qquad \text{where} \qquad -\frac{\partial}{\partial u_\alpha}\left(-\sqrt{u_0^2 - \vec{u}\cdot\vec{u}}\right) = \frac{\gamma u^\alpha}{c}$$

with $\partial/\partial u_\alpha = (\partial/\partial u_0\,,\, -\partial/\partial \vec{u})$. Since $cP^\alpha - qA^\alpha = mcU^\alpha = \gamma mc(c\,,\,\vec{u}\,)$

$$\vec{u} = \frac{c\vec{P} - q\vec{A}}{\gamma mc} \quad \Longrightarrow \quad 1-\beta^2 = 1 - \frac{|c\vec{P} - q\vec{A}\,|^2}{\gamma^2 m^2 c^4} \quad \Longrightarrow \quad \gamma^2 = \frac{|c\vec{P} - q\vec{A}\,|^2 + m^2c^4}{m^2c^4}$$

Therefore,

$$\vec{u} \;=\; \frac{c\vec{P} - q\vec{A}}{\gamma mc} \;=\; \frac{c\vec{P} - q\vec{A}}{\sqrt{|\vec{P} - (q/c)\vec{A}|^2 + m^2c^2}}$$

and the relativistic Hamiltonian is defined as

$$H = \vec{P}\cdot\vec{u} - \mathcal{L} = \sqrt{|c\vec{P} - q\vec{A}\,|^2 + m^2c^4} + q\Phi$$

The Hamilton's equation is again equivalent to the equation of motion in Eq. (1).

3. Solution of Wave Equation in Non-Covariant Form (Textbook Section 6.4)

The electromagnetic wave equations are

$$\begin{cases} \dfrac{1}{c^2}\dfrac{\partial^2 \Phi}{\partial t^2} - \nabla^2\Phi = 4\pi\rho \\[2ex] \dfrac{1}{c^2}\dfrac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2\vec{A} = \dfrac{4\pi}{c}\vec{J} \end{cases} \tag{7}$$

In general, a wave equation is in the form of

$$\frac{1}{c^2}\frac{\partial^2 \Psi}{\partial t^2} - \nabla^2\Psi = 4\pi f(\vec{r}, t) \tag{8}$$

where $f(\vec{r}, t)$ is a known source distribution. To solve this equation, we write Ψ and f into Fourier transformations as

$$\Psi(\vec{r}, t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \Psi(\vec{r}, \omega)e^{-i\omega t}d\omega \qquad \text{and} \qquad f(\vec{r}, t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\vec{r}, \omega)e^{-i\omega t}d\omega \tag{9}$$

and the inverse transformations are

$$\Psi(\vec{r}, \omega) = \int_{-\infty}^{\infty} \Psi(\vec{r}, t)e^{i\omega t}dt \qquad \text{and} \qquad f(\vec{r}, \omega) = \int_{-\infty}^{\infty} f(\vec{r}, t)e^{i\omega t}dt \tag{10}$$

The wave equation in Eq. (8) can then be transferred into inhomogeneous Helmholtz equation,

$$-(k^2 + \nabla^2)\Psi(\vec{r}, \omega) = 4\pi f(\vec{r}, \omega) \tag{11}$$

where $k = \omega/c$. To solve the Helmholtz equation, we use Green's function technique by solving the Green's function in k-space (frequency space) from

$$-(k^2+\nabla^2)G_k(\vec{r},\vec{r}\,') = 4\pi e^{i\omega t'}\delta(\vec{r}-\vec{r}\,') \tag{12}$$

Without boundary surface or with a spherically symmetric boundary around $\vec{r}\,'$, $G_k(\vec{r},\vec{r}\,')$ should be spherically symmetric around $\vec{r}\,'$,

$$G_k(\vec{r},\vec{r}\,') = G_k(|\vec{r}-\vec{r}\,'|)$$

The Green's function in the k-space for the infinite boundary condition, $i.e.$ $G_k(|\vec{r}-\vec{r}\,'|)=0$ when $|\vec{r}-\vec{r}\,'|\to\infty$, is therefore

$$G_k^{(\pm)}(|\vec{r}-\vec{r}\,'|) = \frac{1}{|\vec{r}-\vec{r}\,'|}\,e^{\pm ik|\vec{r}-\vec{r}\,'|}e^{i\omega t'} \tag{13}$$

It is easy to verify this is the solution of Eq. (12) by $\vec{\xi}=\vec{r}-\vec{r}\,'$ and for a system with the spherically symmetry around $\vec{r}\,'$ (or $\vec{\xi}=0$),

$$\begin{aligned}
-(k^2+\nabla^2)G_k(|\vec{r}-\vec{r}\,'|) &= -\left(k^2+\frac{1}{\xi^2}\frac{\partial}{\partial\xi}\,\xi^2\frac{\partial}{\partial\xi}\right)G_k(\xi)\\
&= -\left(k^2+\frac{1}{\xi^2}\frac{\partial}{\partial\xi}\,\xi^2\frac{\partial}{\partial\xi}\right)\frac{1}{\xi}\,e^{\pm ik\xi} \;=\; 0
\end{aligned}$$

The Green's function for the wave equation in Eq. (11) is the inverse Fourier transformation of the solution in Eq. (13),

$$\begin{aligned}
G^{(\pm)}(\vec{r},t,\vec{r}\,',t') &= G^{(\pm)}(|\vec{r}-\vec{r}\,'|\,,t-t') \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{|\vec{r}-\vec{r}\,'|}\,e^{\pm ik|\vec{r}-\vec{r}\,'|}e^{i\omega t'}d\omega\\
&= \frac{1}{|\vec{r}-\vec{r}\,'|}\,\delta\left(t'-t\pm\frac{1}{c}\,|\vec{r}-\vec{r}\,'|\right)
\end{aligned} \tag{14}$$

where G^{+} and G^{-} are the retarded (with causality) and advanced (without causality) Green's function, respectively. For physical situation, we should use the retarded solution with the causality, where

$$t' = t-\frac{1}{c}\,|\vec{r}-\vec{r}\,'|$$

The solution of the inhomogeneous wave equation in Eq. (8) is then

$$\Psi(\vec{r},t) = \Psi_{in}(\vec{r},t)+\int G^{(+)}(|\vec{r}-\vec{r}\,'|\,,t-t')f(\vec{r}\,',t')\,d^3\vec{r}\,'dt' \tag{15}$$

where $\Psi_{in}(\vec{r},t)$ is the solution of homogeneous $[f(\vec{r},t)=0]$ wave equation. If $\Psi_{in}(\vec{r},t)=0$,

$$\begin{aligned}
\Psi(\vec{r},t) &= \int G^{(+)}(|\vec{r}-\vec{r}\,'|\,,t-t')f(\vec{r}\,',t')\,d^3\vec{r}\,'dt'\\
&= \int\frac{1}{|\vec{r}-\vec{r}\,'|}\,f(\vec{r}\,',t')\,\delta\left(t'-t+|\vec{r}-\vec{r}\,'|/c\right)\,d^3\vec{r}\,'dt'\\
&= \int\frac{1}{|\vec{r}-\vec{r}\,'|}\,f\left(\vec{r}\,',\,t-|\vec{r}-\vec{r}\,'|/c\right)\,d^3\vec{r}\,'
\end{aligned}$$

For a single charge moving along a trajectory $\vec{r} = \vec{r}_c(t)$, the charge density can be written as

$$\rho(\vec{r}, t) = q\, \delta(\vec{r} - \vec{r}_c(t))$$

and the electric potential radiated from the charge is then

$$\Phi(\vec{r}, t) = q \int \frac{1}{|\vec{r} - \vec{r}\,'|}\, \delta\left(\vec{r}\,' - \vec{r}_c\left(t - |\vec{r} - \vec{r}\,'|/c\right)\right)\, d^3\vec{r}\,'$$

For a charge moving with a constant velocity, $\vec{r}_c(t) = \vec{v}_0\, t$, we can choose a coordinate such that $\vec{v}_0 \parallel \vec{e}_x$ and then

$$\begin{aligned}
\Phi(\vec{r}, t) \quad &= \quad q \int \frac{1}{|\vec{r} - \vec{r}\,'|}\, \delta\left(\vec{r}\,' - \vec{v}_0 t + \vec{\beta}\, |\vec{r} - \vec{r}\,'|\right) d^3\vec{r}\,' \\
&= \quad q \int \frac{1}{|\vec{r} - \vec{r}\,'|}\, \delta\left(x' - v_0 t + \beta\, |\vec{r} - \vec{r}\,'|\right)\, \delta(y')\, \delta(z')\, dx' dy' dz' \\
&= \quad q \int \frac{1}{\sqrt{(x - x')^2 + y^2 + z^2}}\, \delta\left(x' - v_0 t + \beta \sqrt{(x - x')^2 + y^2 + z^2}\right) dx' \\
&\overset{\xi = x - x'}{=} \quad -q \int \frac{1}{\sqrt{\xi^2 + y^2 + z^2}}\, \delta\left(-\xi + (x - v_0 t) + \beta \sqrt{\xi^2 + y^2 + z^2}\right) d\xi \\
&= \quad \frac{q}{\sqrt{\xi^2 + y^2 + z^2} - \beta \xi}
\end{aligned}$$

where $\xi = x - x' > 0$ is the positive root of

$$-\xi + (x - v_0 t) + \beta \sqrt{\xi^2 + y^2 + z^2} = 0$$

This integral was evaluated using

$$\int F(\xi) \delta(g(\xi)) d\xi = \sum_i \frac{F(\xi_i)}{|g'(\xi_i)|}$$

with

$$g(\xi) = -\xi + (x - v_0 t) + \beta \sqrt{\xi^2 + y^2 + z^2} \qquad \text{and} \qquad g'(\xi) = -\frac{\sqrt{\xi^2 + y^2 + z^2} - \beta \xi}{\sqrt{\xi^2 + y^2 + z^2}}$$

where x_i are the real root of $g(x_i) = 0$.

3. Solution of Wave Equation in Covariant Form —— Invariant Green's Functions

The covariant form of electromagnetic wave equation in 4-dimensional time-space is

$$\partial^\beta \partial_\beta A^\alpha = \frac{4\pi}{c} J^\alpha \tag{16}$$

where $\partial^\alpha = (\partial/\partial x_0\,, \nabla)$, $\partial_\alpha = (\partial/\partial x_0\,, -\nabla)$, $A^\alpha = (\Phi\,, \vec{A})$, and $J^\alpha = (c\rho\,, \vec{J})$. The covariant Green's function for the wave equation in 4-dimensional time-space is

$$\partial^\beta \partial_\beta D(\mathbf{x} - \mathbf{x}') = \delta^{(4)}(\mathbf{x} - \mathbf{x}') \tag{17}$$

where $\delta^{(4)}(\mathbf{x}-\mathbf{x}') = \delta(x_0-x_0')\,\delta^{(3)}(\vec{r}-\vec{r}\,')$ is the 4-dimensional delta-function. With the infinite boundary condition, the solution of the covariant wave equation in Eq. (16) can be calculated from the covariant Green's function as

$$A^\alpha(\mathbf{x}) = A^\alpha_{in}(\mathbf{x}) + \frac{4\pi}{c}\int D(\mathbf{x}-\mathbf{x}')\,J^\alpha(\mathbf{x}')\,d^4\mathbf{x}' \tag{18}$$

where A^α_{in} is the solution of the homogeneous wave equation with $J^\alpha = 0$, which is a background or incident field and not from fields of any charges or currents inside the space domain considered. In the following, we will derive the covariant Green's function $D(\mathbf{x}-\mathbf{x}')$ by converting the 3D Green's function in Eq. (14) into a 4D form. Note that $D(\mathbf{x}-\mathbf{x}')$ defined in Eq. (18) has an extra factor $4\pi/c$ in compared with the 3D Green's function defined in Eq. (15), where $1/c$ is due to an extra $1/c$ factor in the 4D wave equation in Eq. (16) in compared with the 3D wave equation in Eq. (7). We will compensate 4π in the final result of $D(\mathbf{x}-\mathbf{x}')$. Recall the 3D Green's function

$$G^{(\pm)}(\vec{r}-\vec{r}\,', t-t') = \frac{1}{|\vec{r}-\vec{r}\,'|}\,\delta\left(t'-t\pm\frac{1}{c}\,|\vec{r}-\vec{r}\,'|\right) \tag{19}$$

or

$$G^{(\pm)}(R, x_0-x_0') = \frac{1}{R}\,\delta(x_0'-x_0\pm R) \qquad \text{with} \qquad \vec{R} = \vec{r}-\vec{r}\,' \tag{20}$$

Note that $x_0 > x_0'$ for the retarded solution G^+ and $x_0 < x_0'$ for the advanced solution G^-. This 3D Green's function can also be written into a combined form as

$$G^{(\pm)}(R, x_0-x_0') = \frac{\theta\left(\pm(x_0-x_0')\right)}{R}\,\left[\delta(x_0'-x_0+R)+\delta(x_0'-x_0-R)\right] \tag{21}$$

where $\theta(x)$ is the step function, *i.e.* $\theta(x) = 0$ when $x < 0$ and $\theta(x) = 1$ when $x \geq 0$. The use of the step function here is to pick one of the two delta-functions in Eq. (21) for G^+ or G^-. The two delta-functions in Eq. (21) can be combined into a single delta-function in 4D Minkowski space as

$$\begin{aligned}\frac{1}{R}\,\left[\delta(x_0'-x_0+R)+\delta(x_0'-x_0-R)\right] \;&=\; 2\delta\left((x_0'-x_0+R)(x_0'-x_0-R)\right)\\ &=\; 2\delta((x_0'-x_0)^2-|\vec{x}\,'-\vec{x}|^2) \;=\; 2\delta(|\mathbf{x}-\mathbf{x}'|^2)\end{aligned} \tag{22}$$

where $|\mathbf{x}-\mathbf{x}'|^2 = (x_0'-x_0)^2 - |\vec{x}\,'-\vec{x}|^2 = (\mathbf{x}-\mathbf{x}')_\alpha(\mathbf{x}-\mathbf{x}')^\alpha$ is the scalar distance in the Minkowski space. To obtain Eq. (22), we have used

$$\delta(g(y)) = \sum_i \frac{\delta(y-y_i)}{|g'(y_i)|}$$

where y_i are the real roots of $g(y_i) = 0$ and

$$\left\{\begin{array}{rcl} g(y) &=& (y-x_0+R)(y-x_0-R) \\ g(y) &=& (y-x_0+R)(y-x_0-R) \;=\; 0 \qquad \Longrightarrow \qquad y-x_0 = \pm R \\ |g'(y)| &=& 2|(y-x_0)| \;=\; 2R \end{array}\right.$$

The 3D Green's function in Eq. (21) can then be written as

$$G^{(\pm)}(R, x_0 - x_0') = 2\theta\left(\pm(x_0 - x_0')\right)\delta(|\mathbf{x} - \mathbf{x}'|^2)$$

With the consideration of a 4π factor difference in the definition of Green's function in Eqs. (15) and (18), the Green's function in the 4D Minkowski space can be obtained from the Green's function in the 3D Euclidean space as

$$D(\mathbf{x} - \mathbf{x}') = \frac{1}{4\pi}G^{(\pm)}(R, x_0 - x_0') = \frac{1}{2\pi}\theta\left(\pm(x_0 - x_0')\right)\delta(|\mathbf{x} - \mathbf{x}'|^2)$$

In the 4D Minkowski space, the retarded Green's function is therefore

$$D_r(\mathbf{x} - \mathbf{x}') = \frac{1}{2\pi}\theta(x_0 - x_0')\delta(|\mathbf{x} - \mathbf{x}'|^2) \tag{23}$$

and the advanced Green's function is

$$D_a(\mathbf{x} - \mathbf{x}') = \frac{1}{2\pi}\theta(x_0' - x_0)\delta(|\mathbf{x} - \mathbf{x}'|^2) \tag{24}$$

HW. 12.3, 12.5, 12.6, 12.11